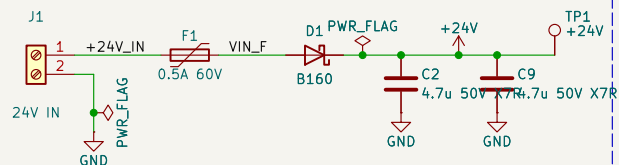
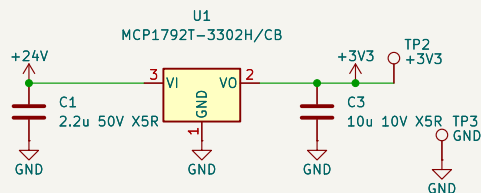


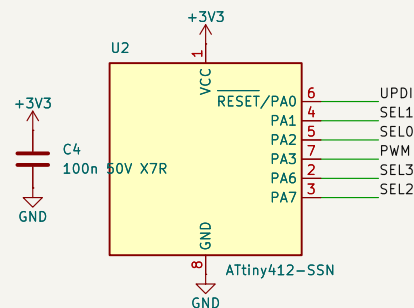
Input & protection



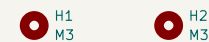
3.3 V regulator



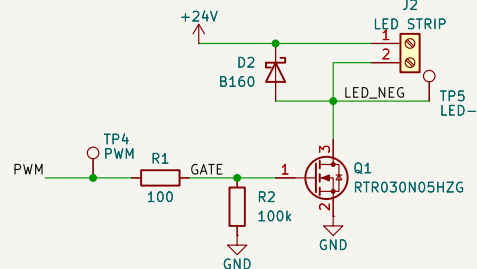
MCU



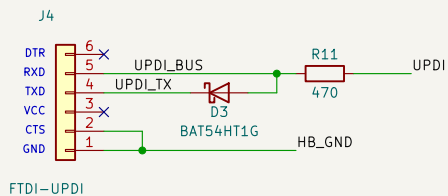
Mechanical



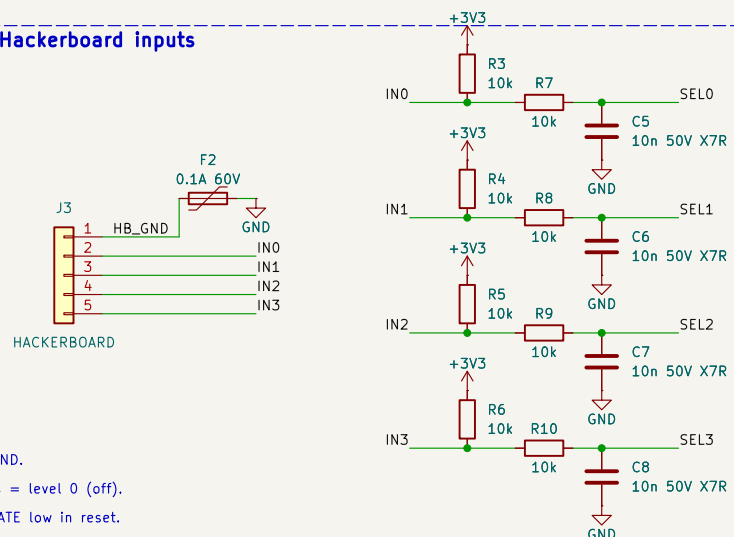
LED output stage



Programming (SerialUPDI)



Hackerboard inputs



Hackerboard pins 0-3 are open-drain: 1 = pulled to GND.

Firmware: level = ~(SEL3..SEL0) & 0x0F, so open inputs = level 0 (off).

PWM: TCA0 W00 on PA3, 10-bit, ~2.44 kHz. R2 holds GATE low in reset.

Flash: 3.3 V USB-serial adapter on J4, board powered from 24 V.

avrdude -c serialupdi -p t412 -P <port> -U flash:w:fw.hex

Level selected by Prusa GPIO Hackerboard pins 0-3
ATtiny412 + N-MOSFET low-side PWM for a 24 V LED strip
hyiger

Sheet: /

File: led-pwm-4bit.kicad_sch

Title: 4-Bit LED PWM Controller

Size: A4

Date: 2026-10-07

Rev: 0.1

KiCad E.D.A. 10.0.7

Id: 1/1